

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Cryptography and Network Security  
**COURSE CODE:** CSA51

### COURSE OUTCOME COVERED

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

**QUESTIONS:** 5

**Total Marks:** 50

Annexure A

### APPLICATION- BASED QUESTIONS

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#### Code of Conduct:

I C. Priyanka / 125107 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Priyanka

95  
100



## Annexure A

### APPLICATION-BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
  - a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
  - b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
  - a. Deduce the most likely original message.
  - b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.
3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
  - a. Demonstrate how the encryption process transforms the message step-by-step.
  - b. Explain how the repeating keyword influences each character of the ciphertext.
4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
  - a. Analyze how investigators could detect the presence of hidden data within the image.
  - b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".
  - a. Determine the most probable original message by reversing the encryption process.
  - b. Critically evaluate why such a method is vulnerable in modern communication systems.
  - c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.



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Course

Code: CSA5102.

### CO1 Assessment Tool 1

Q3) Vigenere Cipher

Given,

Plaintext:  
HELLO

Key:  
KEY

Since the plaintext has 5 Letters and the key has 3 Letters  
repeat the key:

Step 1: Write the Letters

| Position | Plaintext | Key |
|----------|-----------|-----|
| 1        | H         | K   |
| 2        | E         | E   |
| 3        | L         | Y   |
| 4        | L         | K   |
| 5        | O         | E   |

Step 2: Convert letters into numbers

using:

A=0, B=1, C=2, ... Z=25



| Plaintext | Value | Key | Value |
|-----------|-------|-----|-------|
| H         | 7     | K   | 10    |
| E         | 4     | E   | 4     |
| L         | 11    | Y   | 24    |
| L         | 11    | K   | 10    |
| O         | 14    | E   | 4     |

Step 3: Apply encryption formula  

$$\text{Ciphertext} = (\text{Plaintext} + \text{Key}) \bmod 26$$

1.  $H + K \rightarrow 7 + 10 = 17 \rightarrow R$
2.  $E + E \rightarrow 4 + 4 = 8 \rightarrow I$
3.  $L + Y \rightarrow 11 + 24 = 35 \rightarrow 9 \rightarrow J$
4.  $L + K \rightarrow 11 + 10 = 21 \rightarrow V$
5.  $O + E \rightarrow 14 + 4 = 18 \rightarrow S$

HELLO  $\rightarrow$  RIJVS

Conclusion:

The Vignere Cipher Provides Stronger Security than a simple fixed-shift Cipher because different plaintext characters can be encryption using different shifts depending on the keyword.



## Transposition cipher

### Decrypt the message

The question mentions a transposition-based encryption method with three levels of arrangement. This corresponds to a 3-rail Rail Fence transposition.

Given ciphertext:

WECRLTEERDSOEFEADCAIDEN

For a 3-Rail Fence Cipher, the characters are arranged across three rows in a zig-zag pattern.

Original message:

WE ARE DISCOVERED FLEE AT ONCE.

Without spaces:

WEAREDISCOVEREDFLEEFATONCE

So:

Cipher text:

WECRLTEERDSOEFEADCAIDEN

(b) Why is this method vulnerable?

A transposition cipher only changes the positions of characters. It does not fundamentally change the characters themselves.



For examples:

Original:

WEARE---

The same letters are still present in the ciphertext; they have simply been rearranged.

Main weakness:

1. Small Key Space
2. Frequency information is preserved
3. Easy to attack with modern computing
4. Not Suitable for Modern Secure Communication

(c) Practical enhancement

A practical improvement is to replace the simple transposition cipher with a modern symmetric encryption algorithm such as AES.

Why AES?

AES:

- Uses a Secret Key.
- Provides strong Confidentiality.
- Is resistant to practical brute-force attacks when properly configured.
- Is widely used for secure data protection.



There are 2 types in transposition technique.

1. Rail Fence Transposition

2. Columnar Transposition

### 1. Rail Fence Transposition

The Rail Fence cipher is the simplest transposition cipher. The steps to obtain cipher text from this technique are as follows.

Step 1:

The plain text is written as a sequence of diagonals.

Step 2:

then, to obtain the cipher text the text is read as a sequence of rows.

Plain text: MEET ME TOMORROW

Rail 1:

M → E → T → M → E → T → O → R → O → W

MENTRO/ETORW

Rail 3:

M → E → T → M → E → T → O → R → O → W

MMMO/ETEOORW/ETR

## Columnar Transposition

Plaintext : MEET TOMORROW

Key : BEST = 4

|   |   |   |    |    |
|---|---|---|----|----|
|   | 1 | 4 | 18 | 19 |
|   | B | E | S  | T  |
| M | E | E | T  |    |
| T | O | M | O  |    |
| R | R | O | W  |    |

MTR/ EOR/ EMO/ TDW

Next me tomorrow

|   |   |   |    |    |
|---|---|---|----|----|
|   | 1 | 4 | 18 | 19 |
|   | B | E | S  | T  |
| M | E | E | T  |    |
| M | E | T | O  |    |
| M | O | R | R  |    |
| O | W | X | X  |    |

MMMO/ EEOW/ ERRX/ TORX

Next me tomorrow

|   |   |   |    |    |
|---|---|---|----|----|
|   | 1 | 4 | 11 | 15 |
|   | H | E | L  | D  |
| M | E | E | T  |    |
| M | E | T | O  |    |
| M | O | R | R  |    |
| O | W | X | Y  |    |

EEOW/ MMMO/ ETRX/ TORX





# RUBRICS FOR CALCULATION:

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |
| Analysis of Results       | 15        | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results            |
| Documentation             | 10        | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                 |

20

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20

15

10

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100

A